

A WEB-SUPPORTED NATIONAL BRIDGE INVENTORY MANAGEMENT TOOL

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The U.S. National Bridge Inventory (NBI) contains approximately 600,000 bridges. Its size, numerous items, and electronic format make management and analysis of its data difficult. The objective of this study is to develop a tool that will provide a user-friendly environment as well as advanced services for accessing and analyzing NBI data sets. The tool can import text- or spreadsheet-based NBI data sets, provide tools for managing and analyzing them, and use Web services for pinpointing bridge positions. It is considered useful for managing the NBI data sets and for further examining and analyzing the characteristics and condition of the NBI.

Keywords: bridge; database; multimedia; Internet; National Bridge Inventory

The National Bridge Inventory (NBI) contains data on about 600,000 state and local bridges in the United States (Chase, Small, & Nutakor, 2000; U.S. Federal Highway Administration [FHWA], 2003). These data are gathered periodically from state and local agencies and submitted to the U.S. Department of Transportation. Agencies are required to provide the data in an electronic form (either in ASCII or Microsoft Excel) and in a specific format described in the *Recording and Coding Guide for the Structure Inventory and Appraisal of the Nation's Bridges* (FHWA, 1995). The ASCII format of the data can be considered out of date and difficult to use because it dates from the early 1980s. The FHWA (2003) recently published NBI data sets in Microsoft Excel format on the Web.

The objective of the present study is to develop a tool that will aid engineers and other professionals in accessing, retrieving, manipulating, or obtaining information from NBI data. This tool is based on commonly used software—Microsoft Access—is strictly academic, and can be freely used by any interested party with the appropriate acknowledgement. It purports not to replace existing bridge management system (BMS) databases but rather, to provide an analysis utility independent of any specific NBI data set. The remainder of this article is organized as follows. The second section briefly describes the NBI and provides background on recent studies related to the subject. The third and fourth sections provide an overview and description of the capabilities of the developed tool. A demonstration of the tool is described in the fifth section. Finally, the sixth section summarizes the conclusions of the study.

Background

The NBI, initiated in 1972, is a data set containing information about state, local, park, railroad, and private bridges in the United States. The NBI contains more than 150 information

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fields or items for each bridge, including the type, material, construction characteristics, position of the bridge, assessments on its condition (condition indices), traffic information, past actions on the bridge, and proposed actions for bridge maintenance. Each item has a specific coding structure, which is described by the FHWA (1995). The NBI bridges are under the authority of state departments of transportation, local agencies, and park agencies. Data in the NBI are collected in annual or 2-year periods by Department of Transportation teams. Items not specified during construction, such as condition indices, are gathered based on site inspections (FHWA, 1995; Tonias, 1994). Data not directly related to bridges (e.g., traffic characteristics) are extracted from other databases usually managed by Departments of Transportation.

The NBI is used for the operation of BMSs in the United States. BMSs such as the Indiana Bridge Management System (Sinha, Zhang, & Woods, 2000), Pontis Bridge Management System (Thompson, 1993), and other state or commercial systems need complete or partial NBI information to proceed to their analysis. For that reason, databases have been developed to support such systems. In any case, it is a fact that a BMS's function and performance is widely based on the speed and storage capabilities of its database, like any other software package of that type (Atzeni, Ceri, Paraboschi, & Torlone, 1999). The efficiency of such a database in that case is crucial for the operation of the BMS. As Khan (2000) suggested, the future in development of BMS databases is based on new techniques to access and analyze existing bridge databases. The present study does not focus on such databases to support a BMS—despite the fact that the form and design of such databases are similar.

The tool described in the present study is intended to be used for manipulating NBI database files and retrieving data, both offline and online by researchers or professionals, independently of a BMS software package or an NBI data set. Such tools, not necessarily used by a BMS or focused on bridges, have and are being developed worldwide. Reich, Fenves, and Subrahmanian (1994) proposed a machine-learning technology to support database information extraction. Li (1998) developed a multimedia database for a highway information system, including networking capabilities and a video server. Nambisan et al. (1999) proposed prototype Internet applications for transportation system management to implement in Clark County, Nevada, including Web-supported tools such as a database and Geographic Information System (GIS); the proposed system could provide, in graphical form, maps of areas, transportation characteristics, and so forth with the aid of GIS and was intended to support the transportation system of the county. Alam, Melchiorre, and Clayton (1999) developed a set of Intranet/Internet applications for the Dynamic Presentation of a Cross-Jurisdictional Transportation database. Their focus creates a database by integrating information from various authorities and agencies responsible for parts of the transportation network and provides information to the Intranet and the Internet in a dynamic manner. Wang, Zhang, and Li (2000) built an Internet-based multimedia highway information system, a database containing both engineering and photographic data, with an objective to provide the available information both to an internal highway network and the Web.

Regarding the NBI, Mohammed, Haque, and Pongponrat (2000) constructed a multimedia information system that integrates the Uniform Bridge Element Identification System database for bridge inspection and maintenance. The system could store bridge information and also photos, videos, and so on for supporting bridge inspection and maintenance. The Uniform Bridge Element Identification System number, appointed to each bridge by the federal Department of Transportation, was used by the software to retrieve bridges. Chase et al. (2000) published their study of an in-depth analysis of the NBI using data mining, GIS, and advanced statistical methods. Mining methods, proven to be very efficient in data extraction and descriptive statistics, were used to provide interesting information about the NBI. The analysis was supported by a GIS.

Recently, KwikSoft Development Company developed a database package titled Bridge Information System to accompany specific NBI data sets (National Bridge Inventory, 2002). The interface has multimedia capabilities, can cooperate with modern GIS software packages such as MapPoint and ARCView, and is a commercial package sold along with NBI data for a

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specific state. The package has the capability of giving information (specific or aggregated) for the included data set only. Stidger (2002) provided a description of the package. Also, as a part of a project funded by the FHWA Bridge Management Information Systems Laboratory, GIS/Trans Ltd. is currently developing a spatially enabled Web site that provides Web-accessible tools to the NBI. Such tools include queries and dynamic mapping (Nutakor & Viswanathan, 2003). Finally, Svirsky (2004) developed a World Wide Web portal providing information on bridges for New England and the state of New York based on the 2001 NBI.

Overview

OBJECTIVES AND FRAMEWORK

The tool can be used to manage NBI data and get information about a bridge or a set of bridges, to locate any bridge without necessary GIS and mapping equipment, and to extract partial NBI data sets.

The tool is developed for use as a management utility for NBI data sets, independently of any specific data set or BMS. The objective seeks to transform NBI data sets into user-friendly database tables, providing at the same time enhanced capabilities to the tool user. Such capabilities include retrieving a specific bridge from the inventory, locating a bridge on the map with the help of Web services, running queries, and storing photos of bridges or maps. The tool can be used to manage NBI data and get information about a bridge or a set of bridges, to locate any bridge without necessary GIS and mapping equipment, and to extract partial NBI data sets. The tool is independent of any specific NBI data set, following only the NBI format to import data for use with a state data set, a county data set, or a part of them. Furthermore, it is free, cost effective (solely based on Microsoft Access and the Internet), and easily accessible and expandable.

The tool is basically a database enhanced with a set of procedures that aid in importing and managing NBI data sets as well as providing graphs and reports on statistics for these data sets. It consists of a set of tables, each one corresponding to an imported NBI data set and a number of procedures such as data set importing, data set management and exploring, graph and report production, and Web access. Each table can correspond to any subset of the NBI (such as a state, a county subset, or even subsets of different time periods for the same state). Multiple tables can be loaded. In the tool framework shown in Figure 1, each table is a repository of an NBI data set. The tool can actually handle several NBI data sets at the same time, limited only by hardware storage space and Microsoft Access capabilities. A graphical user interface, consisting of a number of forms, accesses table data and procedures.

PLATFORM USED

The most common software packages for building databases and database-oriented tools are Microsoft Access, Oracle, SQL Server, and the older Clipper and Dbase. The software selected was Microsoft Access. Although other software packages, such as Oracle or SQL Server, are more sophisticated and powerful, Microsoft Access has the following advantages that are considered appropriate for developing such a tool:

- a widespread software package installed on most computers worldwide
- user friendly, easily expandable, and programmable using Visual Basic for Applications
- compatible to other common software packages such as Microsoft Excel
- fully Internet compatible

Microsoft Access 2000 is an easy-to-use, popular database platform with enhanced programming capabilities (Cox, Dudley, & Aune, 1999), which makes it an appropriate platform for the development of the tool.

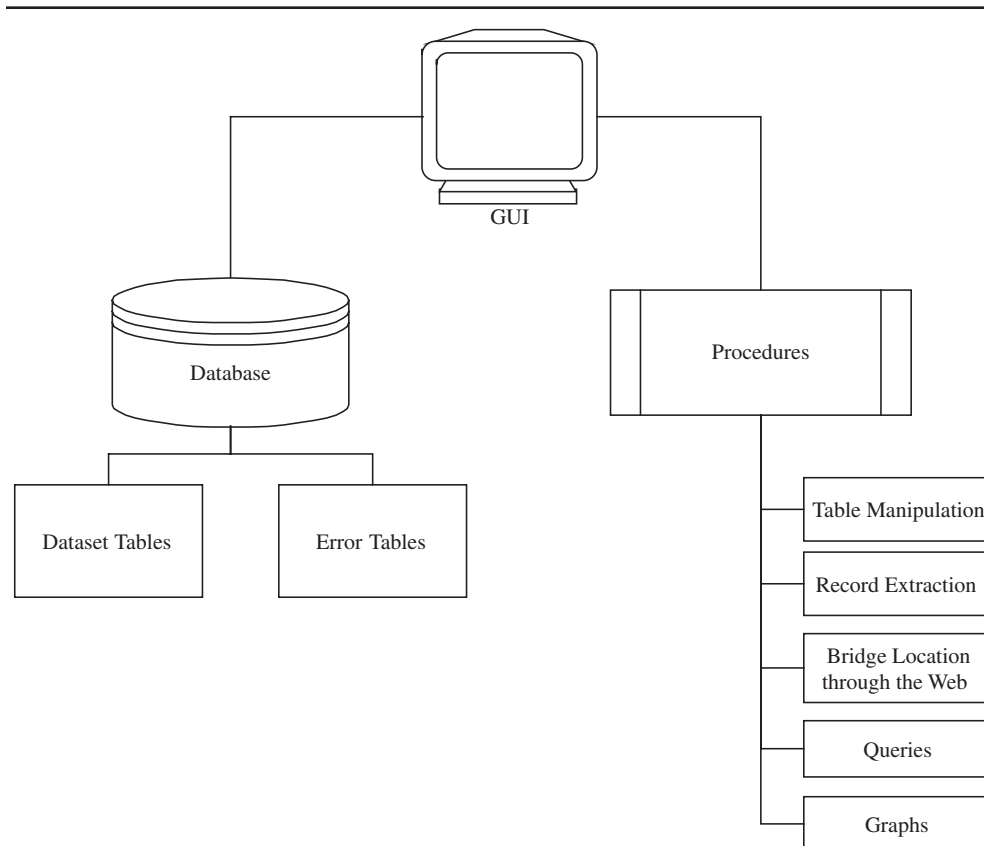


Figure 1: Tool Framework

NOTE: GUI = graphical user interface.

Description

As mentioned earlier, the tool provides a set of procedures for managing and manipulating NBI data sets. These procedures include

- Table manipulation
- Record extraction view
- Map location
- Queries
- Graphs
- Multimedia item storage (photo archive)

The user can manipulate tables containing NBI data sets, browse through table records, find a specific bridge in a table (e.g., using the structure number or the coordinates of a bridge), run queries to get specific information about a group of bridges, obtain graphs on statistics related to NBI items, locate bridges in maps through the Web, and store multimedia information about bridges when available. The procedures, selected from the menu shown in Figure 2, are described in the following subsections.

TABLE MANIPULATION

This is a set of procedures that aid in managing NBI data set tables. The user can import a data set to a table (the data set can be in a ASCII or Microsoft Excel file format), check a table in

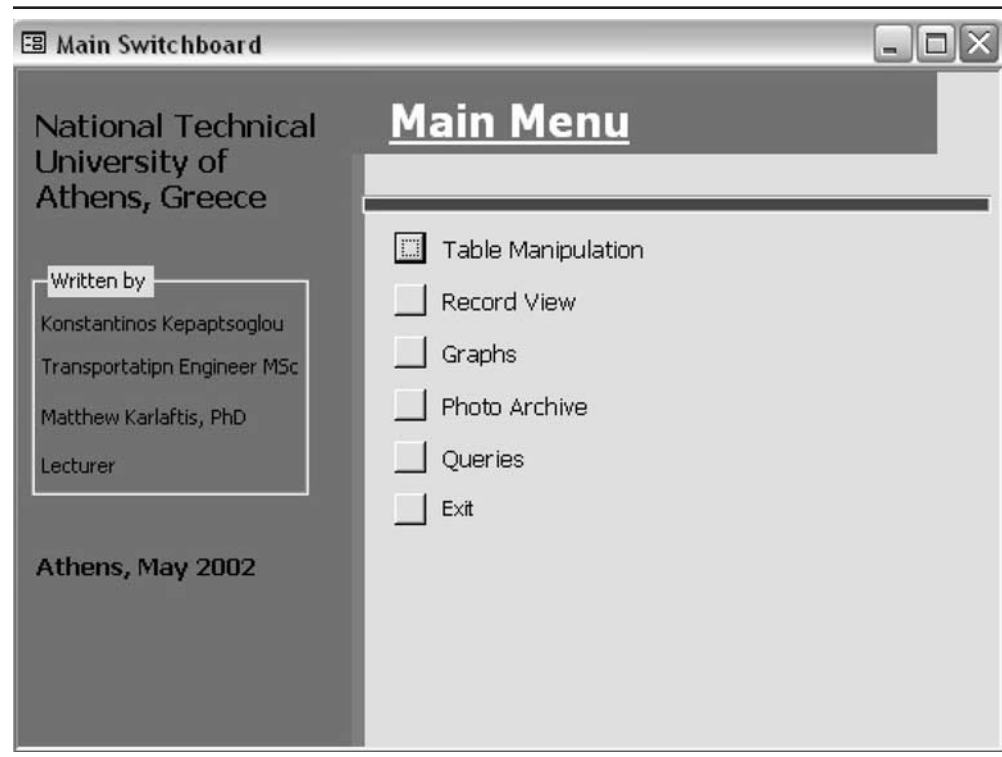


Figure 2: Procedure Selection Menu

data-sheet mode, delete, rename a table, or examine error tables. Tables are imported using the NBI format described by the FHWA (1995). While importing an ASCII file format data set, error tables are produced when items of one or more bridge records are not compatible to the NBI FHWA format. These nonstandard records are filtered to error tables to allow further examination. In addition, the user can export a table in Microsoft Excel format.

RECORD EXTRACTION

This procedure shows a form containing the items of a record for a specific table (see Figure 3). Through the form's options, the user can print the form, manipulate the table's data, search the table for records that contain a specified value for any of the table's items, or locate the bridge shown in the form through the Web. Web services are used to locate a bridge on a map. Because the NBI provides longitude and latitude for each bridge, the tool uses MapQuest Web mapping services (<http://www.mapquest.com>) to pinpoint a bridge on a map. MapQuest is the Web's most popular source for maps and driving directions (MapQuest, 2003). The tool sends longitude and latitude data to the MapQuest Web site and through a Web browser, loads a map with the bridge noted on it.

QUERIES

Queries provide advanced search and partial data set extraction capabilities to a database. The tool allows for the use of queries to obtain insights on parts of an NBI data set. One example is the number of bridges longer than a specific length or deck condition. The user may need limited knowledge of SQL language to use complicated queries. The form used to run a query is shown in Figure 4.

Figure 3: Bridge Record Form (National Bridge Inventory Data Set for Indiana, United States)

Figure 4: Query Example, List of Bridges Built After 1976 (National Bridge Inventory Data Set for Indiana, United States)

GRAPHS

Graphs showing the distribution of bridges for a number of NBI data fields can be produced. Such items include bridge condition indices, age, functional class, design load, operating and inventory rating, material, deck type, and superstructure type. Users can modify the tool to add more graphs according to their needs.



Figure 5: Locating Bridge 3000660 (National Bridge Inventory Data Set for Indiana, United States)

MULTIMEDIA ITEM STORAGE

When photos or maps of bridges are available, they are stored in a separate table related to the table belonging to the bridge and using the state code and structure number of the bridge as primary keys to distinguish the bridge.

Application

A demonstration of the tool is provided for the 1999 NBI data set of Indiana. This data set has 38,454 bridges. Figure 3 shows an example bridge record. Figures 5 and 6 show the locations of Bridge 3000660 and Bridge 6300103, respectively. A benefit of the Web's ability to locate bridges is that it verifies the correctness of the coordinates for a bridge. For example, the bridge shown in Figure 7 (Bridge 41290) should have been on Interstate 70 according to the item Facility Carried by Structure, although on the map it is positioned south of that interstate.

Figures 8 and 9 show the condition distribution for the deck and the superstructure of the data set. As can be seen from both graphs, most bridges are in a condition of 7 or greater, whereas only about 1,000 bridges are in a perfect condition of 9. Figure 10 shows the age distribution of bridges. Most bridges in Indiana were built after the early 1960s. A query could reveal the number of bridges—20,882—that are more than 25 years old, the typical time period for the first bridge rehabilitation.

Another query could provide the bridges on Interstate 65 (see Figure 11). The application shows that the tool can provide several potentials of analysis for the NBI data set without the need of using sophisticated data analysis software packages.

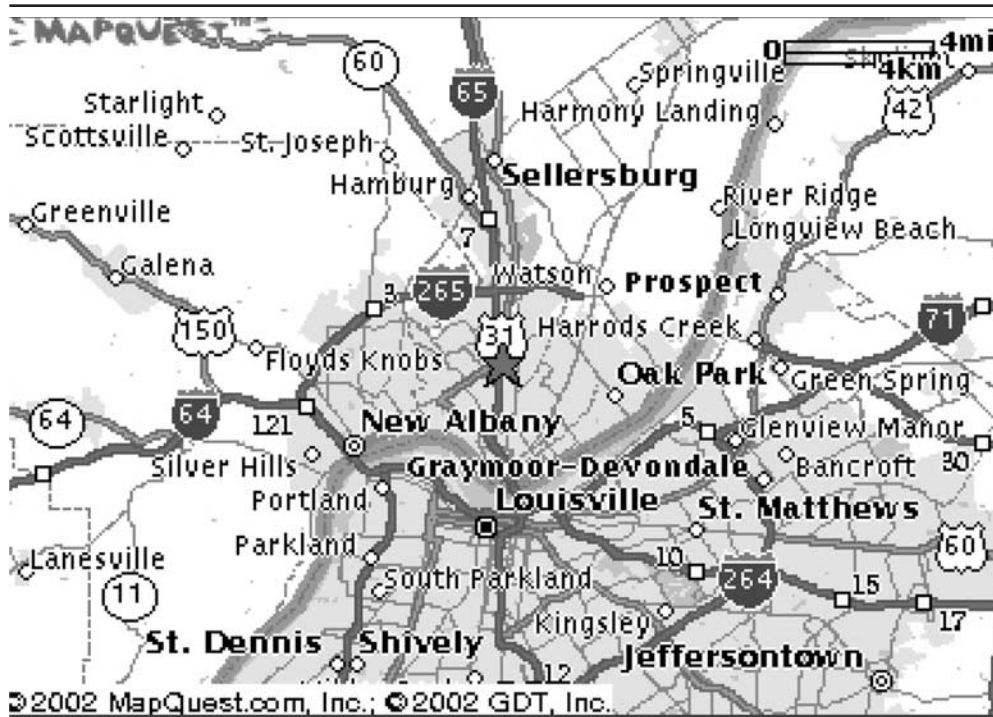


Figure 6: Locating Bridge 6300103 (National Bridge Inventory Data Set for Indiana, United States)

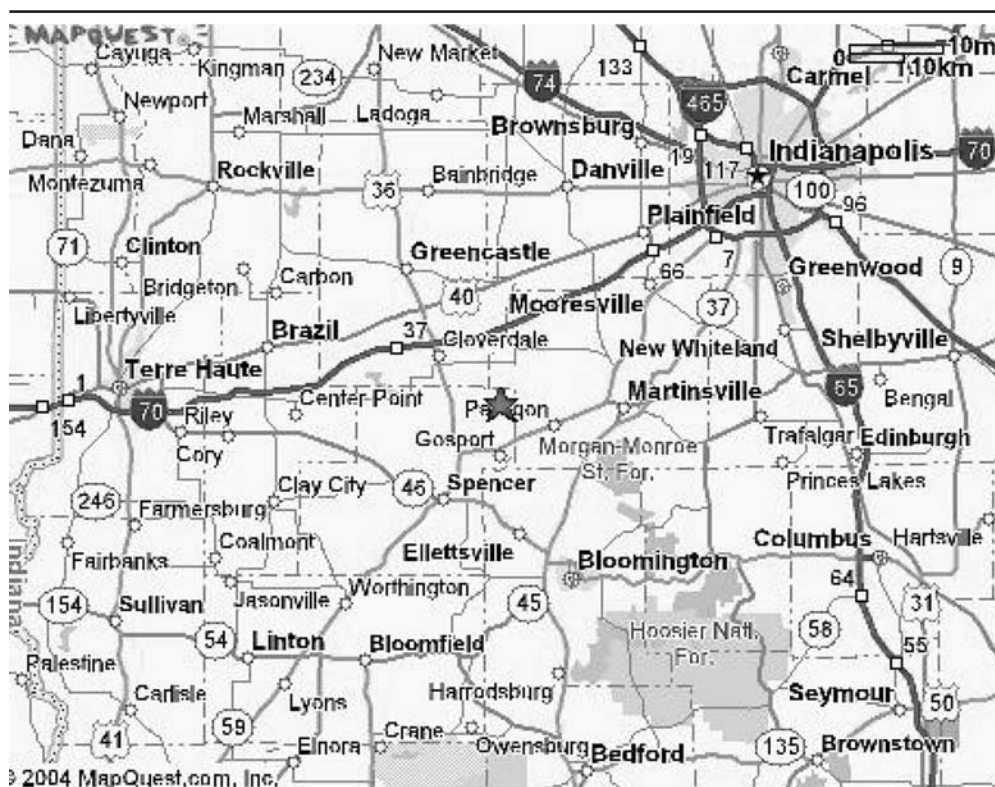


Figure 7: Error in Coordinates of Bridge 41290 (National Bridge Inventory Data Set for Indiana, United States)

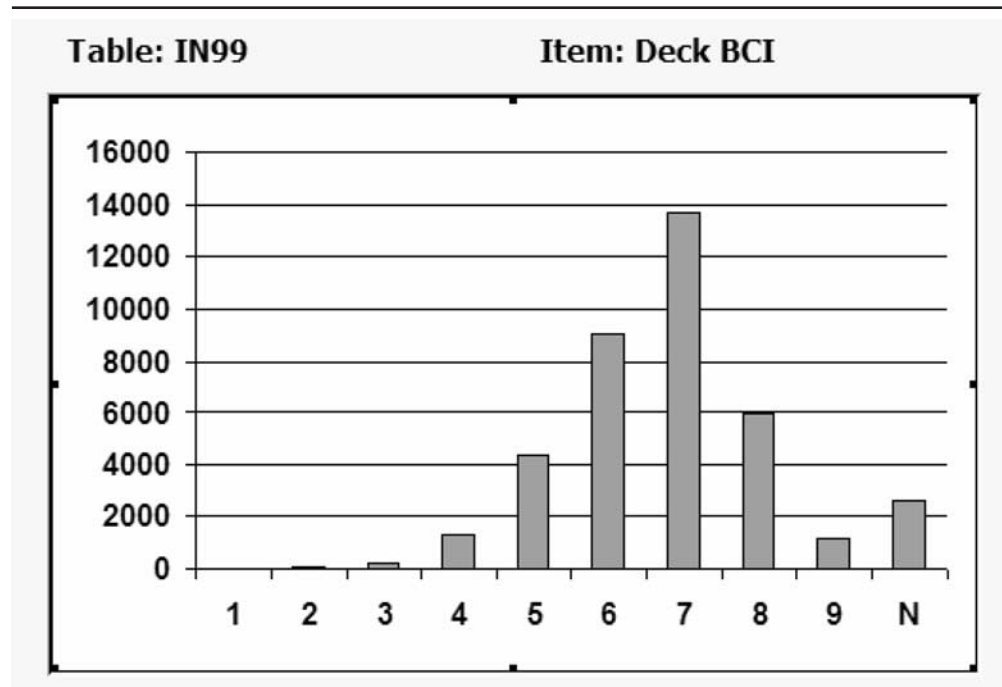


Figure 8: Deck Condition Index Distribution Example Graph (National Bridge Inventory Data Set for Indiana, United States)

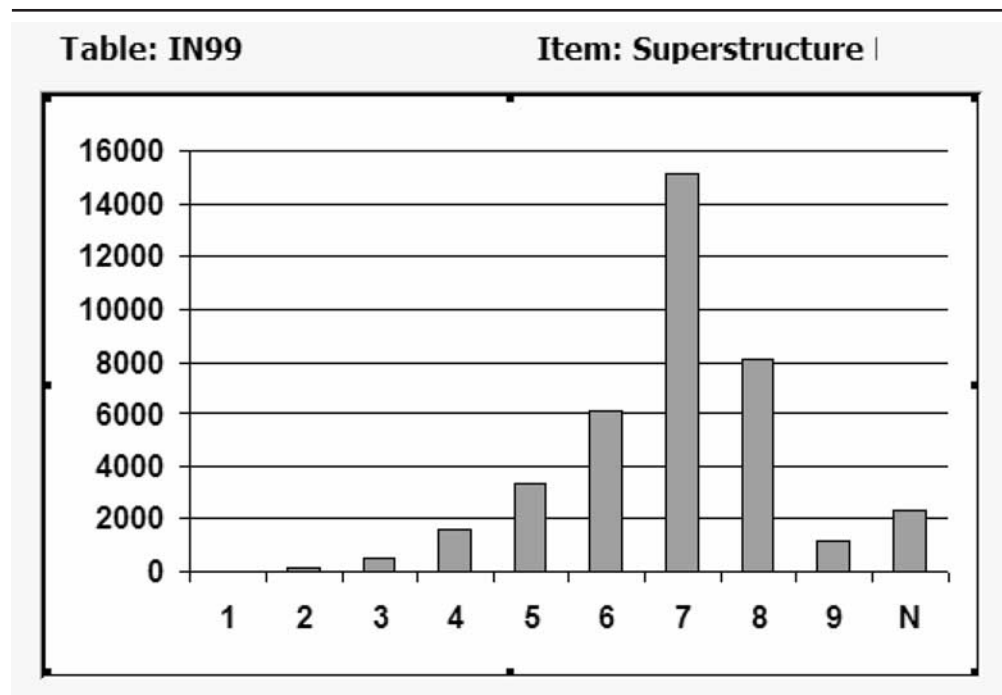


Figure 9: Superstructure Condition Index Distribution Example Graph (National Bridge Inventory Data Set for Indiana, United States)

tistics for the NBI data sets, and provisions for more Web services would be useful. Overall, the tool is considered a cost-effective, user-friendly and most important, fully expandable way of managing NBI data sets.

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